

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CE) Examination

April-May 2013

CE305 Computer Network & Internetworking Layers

Date: 30.04.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following questions.

[06]

1. Explain Service and Interface with diagram.

[02]

2. Explain 1 bit Sliding window protocol.

[02]

3. Describe Optimality Principle and give example.

[02]

(b) Explain Link State Routing Protocol in detail.

[05]

Q - 2 (a) Explain ATM reference model with diagram.

[04]

(b) Explain Selective Repeat ARQ.

[04]

(c) Calculate CRC for the following.

[04]

Frame : $x^9 + x^8 + x^6 + x^4 + x^3 + x + 1$

Generator Polynomial : $x^4 + x + 1$

OR

Q - 2 (a) Explain Go Back N ARQ.

[04]

(b) Differentiate following.

[04]

1. Frame Relay and X.25.

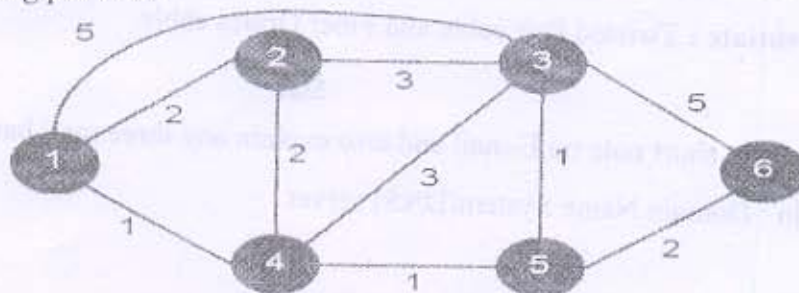
[02]

2. IPV4 and IPV6.

[02]

(c) Draw the Routing Table for **any four** nodes for following topology using Distance Vector Routing protocol.

[04]



Q - 3 (a) Answer the following questions.

1. What is Tunneling? Explain it with suitable example.

2. Explain DQDB.

(b) Draw the figure for IPV4 Header format and explain each term in brief.

(c) Explain : Routing for Mobile Host.

OR

(c) Explain HDLC frame format and also explain the control fields in detail.

SECTION – II

Q - 4 (a) Do as directed

1. Represent 1001101111 in Manchester and Differential Manchester encoding.

2. **Differentiate** : Connection Oriented and Connection Less services.

(b) Answer the following questions.

1. Explain RTS and CTS with example.

2. Explain transport layer Quality of Service parameters.

Q - 5 (a) Which protocols are used as collision free protocols? Explain Bit-Map protocol.

(b) Do as Directed.

1. Draw the Frame format for Ethernet and explain each term in detail.

2. List out Socket Primitives for TCP.

OR

Q - 5 (a) Explain Upward Multiplexing and Downward Multiplexing.

(b) Explain TCP Connection Management in normal case and when collision occurs.

(c) What is DHCP? Why is it used?

Q - 6 (a) What is the main purpose of MAC layer? Explain CSMA/CD.

(b) Explain Two army problem.

(c) Differentiate : Twisted Pair cable and Fiber Optics cable.

OR

Q - 6 (a) Write down short note on E-mail and also explain any three mail handling commands.

(b) Explain : Domain Name System(DNS) server.
